

Thesis Topic Registration Form

Student's Data:

Student's Name: Md Farid

Student's Neptun code: WEFKHB

Educational Information:

Training programme: Computer Science BSc

I have an internal supervisor

Internal Supervisor's Name: *Gnettata walid*

Supervisor's Home Institution: *Department of Artificial Intelligence*

Address of Supervisor's Home Institution: *1117, Budapest, Pázmány Péter sétány 1/C.*

Supervisor's Position and Degree: *PhD Candidate*

Thesis Title: Smart Parking License Plate Recognition & Dynamic Pricing System

Topic of the Thesis:

(Upon consulting with your supervisor, give a 150-300-word-long synopsis of your planned thesis.)

Problem to be solved:

Urban parking management relies heavily on manual processes and fixed pricing, making enforcement inefficient and prone to errors. Current systems often lack automation for vehicle identification, dynamic fee calculation, and administrative control, leading to poor user experience and operational inefficiencies.

Motivation:

Automating license plate recognition and implementing dynamic pricing can improve efficiency, fairness, and revenue optimization. Using pre-trained models allows rapid prototyping of a complete system that combines computer vision, business logic, and a web interface, giving practical experience in integrating AI components without complex model training.

Method of operation:

The system will use pre-trained YOLO for license plate detection and EasyOCR for text recognition. A synthetic dataset will simulate parking zones, sessions, and vehicle registries. The fee engine calculates dynamic rates based on zone, time, and repeat offenses, while all data is persisted in a relational database.

Workflow:

Image upload → plate detection → OCR → plate validation → lookup plate history → dynamic fee calculation → store session → display results via web UI → admin

monitoring and reporting.

Implementation:

A web-based application with a frontend (React/Streamlit) and a Python backend will demonstrate the pipeline. The system includes plate recognition, fee calculation, database operations, and an admin dashboard. Evaluation will measure detection accuracy, fee calculation correctness, response time, and usability.

Output:

The project will deliver a functional web application, synthetic dataset and generation scripts, technical documentation, evaluation results, and a thesis report summarizing implementation, findings, and recommendations for future real-world deployment.

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